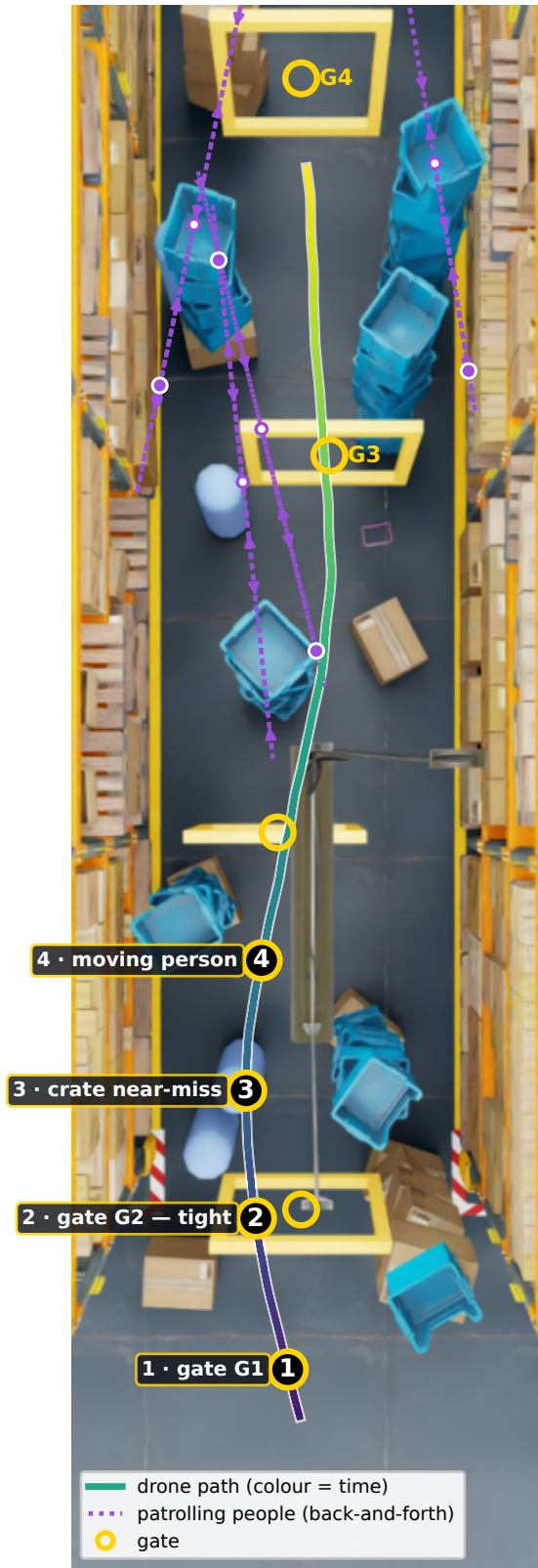
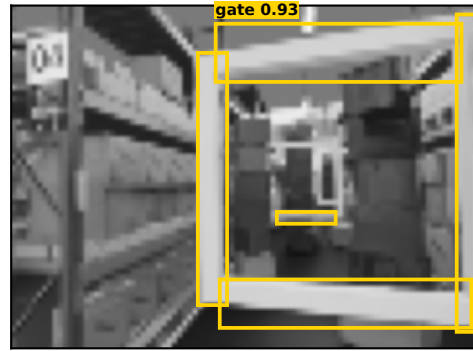


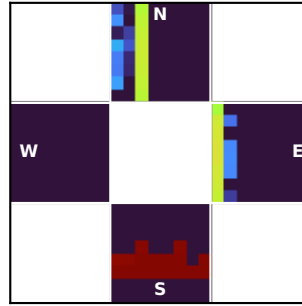
Top-down aisle flight path + moving obstacles



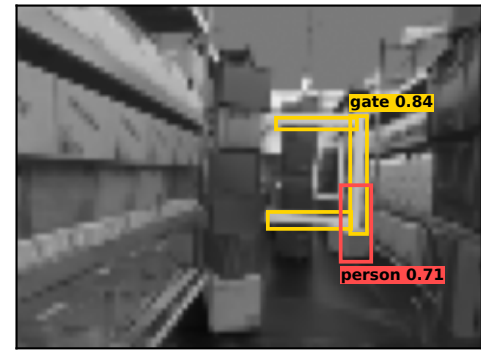
1. gate G1 · $t=2.6s \cdot 0.09\text{ m}$



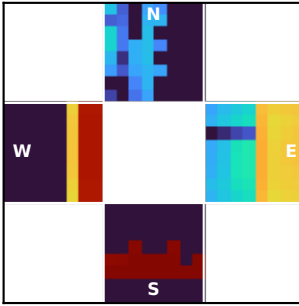
cross-ToF (near=red)



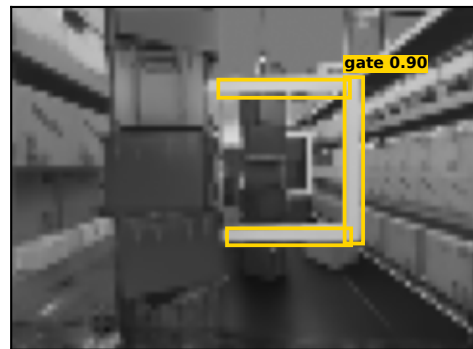
2. gate G2 — tight · $t=5.6s \cdot 0.06\text{ m}$



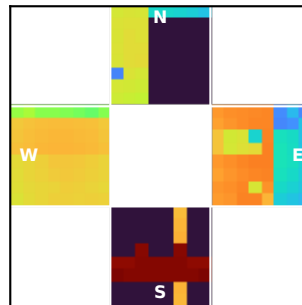
cross-ToF (near=red)



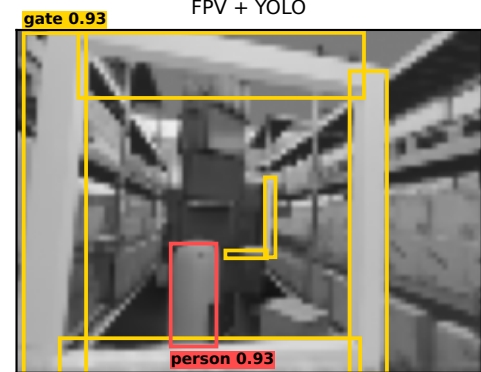
3. crate near-miss · $t=8.0s \cdot 0.66\text{ m}$



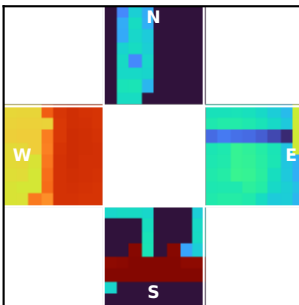
cross-ToF (near=red)



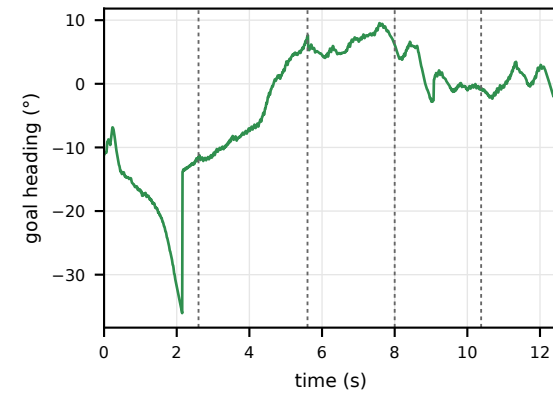
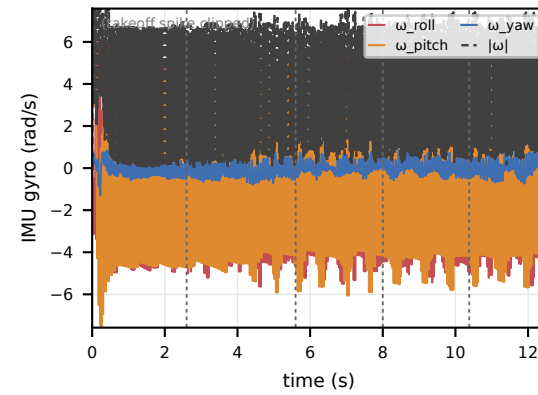
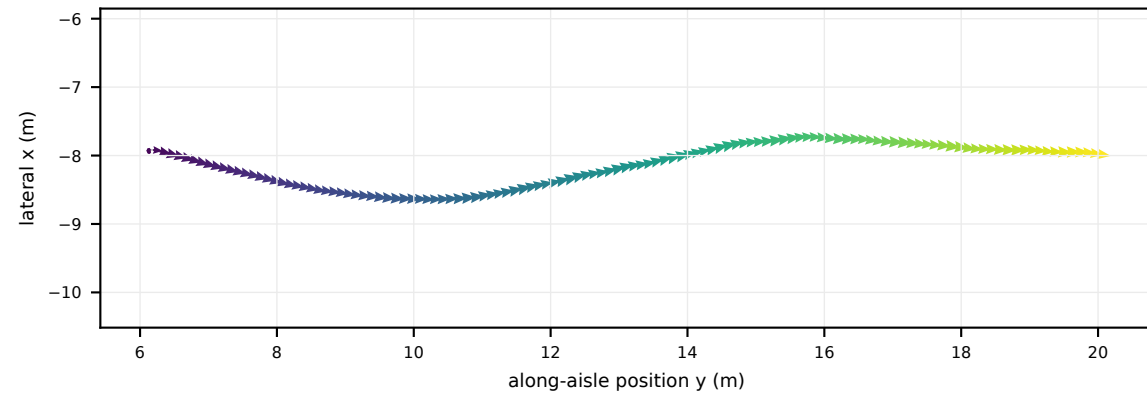
4. moving person · $t=10.4s \cdot 0.61\text{ m}$



cross-ToF (near=red)



velocity — arrow direction = heading, length = forward speed (peak 1.31 m/s)



Onboard K1 warehouse schedule — CP-SAT balances CTRL 100Hz + NAV 50Hz + YOLO sharded across 8 cores (max hart 24ms vs greedy 38ms) · red = sensor in, coloured = output

